

Statistical rigor: is each stage's gain real, or inside the OOF estimation noise?

Question. The tree-search pipeline reports a "ladder" of cross-validated (out-of-fold, OOF) scores per competition — stage 2 (base 3-way blend) → stage 3 (tuned / seed-bagged blend, in rounds $r_1 \rightarrow r_2 \rightarrow r_3$) → stage 4 (tree-search mega-blend winner). Many per-step gains are tiny (0.00x%). This document quantifies, for every stage transition, whether the improvement is **larger than the sampling noise of the OOF estimate itself** — i.e. whether a 95% paired-bootstrap confidence interval on the metric difference excludes 0 (significant) or contains 0 (indistinguishable from noise).

This directly answers the recurring challenge "is that 0.0x% just noise?" — and the honest answer turns out to be **"the whole-ladder gain is real everywhere, but many individual rungs are not individually resolvable above OOF noise, especially in the smaller competitions."**

Generated by `docs/scripts/bootstrap_ci.py` (`uv run python3 docs/scripts/bootstrap_ci.py`). Read-only over `data/` and cached OOF. Nothing retrained, nothing committed.

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Method — paired bootstrap, $B = 5000$

- Pairing.** All stages of one competition share the same cached OOF row order (train file order, identical `KFold(5, shuffle=True, seed=42)` across every stage), so their predictions are aligned row-for-row. The stage-4 winner OOF is reconstructed from the tree winner node's `members` + `dirichlet weights` (`experiments_tree_v3.json` → `node_results`) by weighting the cached member OOFs.
- Gate (pre-flight).** Before bootstrapping, every stage's unresampled metric is recomputed and asserted equal to the recorded ladder value in `competitions/<comp>/experiments.json` (tolerance $1e-5$; recorded to 6 dp). All 25 stage OOFs ($5 \text{ comps} \times 5 \text{ stages}$) reproduced their recorded value to $|\text{diff}| \lesssim 5e-7$ — including the from-scratch stage-4 reconstructions. **No OOF mapped to the wrong stage.**

3. **Resampling.** Each of $B = 5000$ iterations draws one resample of row indices (with replacement, size n) and recomputes every stage's metric on that same resample. Using one shared resample per iteration preserves the (very high) between-stage correlation — this is what makes a paired test powerful for near-identical blends.
4. **Improvement Δ .** For a transition before→after we collect the signed improvement (`after - before` for maximized metrics; `before - after` for RMSE) so improvement is always positive, and report: the point Δ (full data), the 95% percentile CI [2.5, 97.5] of the bootstrap Δ , whether the CI contains 0, and `P(improve)` = fraction of bootstrap draws with $\Delta > 0$.
5. **Metrics** are recomputed with each competition's exact post-processing: RMSE clipped to [0,1]; R^2 clipped to [0,100]; ROC-AUC; accuracy at the best swept threshold (0.10..0.90 step 0.01, plus 0.5) — the accuracy threshold is re-optimized on each resample, so its CI also absorbs threshold-selection variability. Fast count-weighted evaluators are used (validated to reproduce the reference `features.py` /sklearn metric to $\approx 1e-8$, 10–80× faster) so the full run finishes in minutes.

Direction / metric per competition: S5E10 RMSE (minimize), S6E1 R^2 , S6E2 AUC, S4E1 AUC, S4E11 accuracy (all maximize). Δ and CI below are in **units of 10^{-4} metric points**.

Scope limit — read before quoting any of this. This bootstrap quantifies **only** the sampling variability of the OOF estimate on the fixed training rows ("had we drawn a slightly different validation set, how much would this gap wobble?"). It does **not** capture the variance of re-running the whole search with a different seed / fold split / Optuna draw (that needs retraining, deliberately not done here). A gap can be bootstrap-significant here and still be seed-fragile under a full re-run. **Treat these CIs as a lower bound on each gain's uncertainty, not the whole story.** Stage-1 (the February baseline) OOF is not cached, so the 1→2 transition is omitted; the ladder analysed here starts at stage 2.

Headline result — the three canonical rungs

Verdict = whether the 95% CI excludes 0.

Competition	n	2→3	3→4 (tree search)	2→4 (whole ladder)
S5E10 road-accident (RMSE)	517,754	significant	significant	significant
S6E1 exam score (R^2)	630,000	significant	significant	significant
S6E2 heart disease (AUC)	630,000	significant	significant	significant
S4E1 bank churn (AUC)	165,034	significant	inside noise	significant
S4E11 depression (accuracy)	140,700	significant	inside noise	significant

Two clean take-aways:

- **The whole-ladder gain (2→4) is significant in all five competitions.** The full pipeline's improvement over its own base blend is beyond OOF sampling noise everywhere — even though in relative terms every one of these gains is tiny (0.01%–0.11%, see below).

- **The tree-search rung (3→4) is significant only in the three large-n competitions** and falls **inside OOF noise** in the two smaller ones (S4E1 $n \approx 165k$, S4E11 $n \approx 141k$). The tree search's marginal contribution over an already-tuned stage-3 blend is not universally distinguishable from noise; its detectability tracks the number of validation rows.

Detailed CIs (units of 10^{-4} metric points)

Comp	transition	before→after (raw)	Δ	95% CI	P(improve)	verdict
S5E10	2→3	0.056027→0.055976	+0.513	[0.426, 0.598]	1.000	significant
S5E10	3→4	0.055976→0.055968	+0.086	[0.044, 0.126]	1.000	significant
S5E10	2→4	0.056027→0.055968	+0.598	[0.501, 0.694]	1.000	significant
S6E1	2→3	0.786793→0.787060	+2.672	[2.328, 3.030]	1.000	significant
S6E1	3→4	0.787060→0.787183	+1.235	[0.902, 1.554]	1.000	significant
S6E1	2→4	0.786793→0.787183	+3.907	[3.419, 4.417]	1.000	significant
S6E2	2→3	0.955197→0.955510	+3.133	[2.788, 3.477]	1.000	significant
S6E2	3→4	0.955510→0.955529	+0.190	[0.102, 0.277]	1.000	significant
S6E2	2→4	0.955197→0.955529	+3.322	[2.969, 3.669]	1.000	significant
S4E1	2→3	0.894302→0.894354	+0.519	[0.061, 0.975]	0.987	significant
S4E1	3→4	0.894354→0.894393	+0.391	[-0.259, 1.080]	0.883	inside noise
S4E1	2→4	0.894302→0.894393	+0.910	[0.037, 1.809]	0.980	significant (fragile)
S4E11	2→3	0.939488→0.939900	+4.122	[0.498, 7.605]	0.988	significant
S4E11	3→4	0.939900→0.940235	+3.340	[-0.213, 6.041]	0.967	inside noise
S4E11	2→4	0.939488→0.940235	+7.463	[3.127, 10.945]	1.000	significant

Relative size of the total ladder gain (2→4), for calibration against the "0.0x%" framing: S5E10 **0.107%** RMSE reduction • S6E1 **0.050%** R^2 • S6E2 **0.035%** AUC • S4E1 **0.010%** AUC • S4E11 **0.079%** accuracy. All are genuinely "0.0x%" gains — yet all clear OOF noise, because n is large. The statistic that matters is not the raw gap but the gap relative to its CI width.

Fine-grained — is each individual tuning round real? (the sharpest honesty check)

Breaking stage 3 into its rounds (r1, r2, r3) and the final tree-search step exposes how much of the ladder is carried by a few steps while others are pure noise. Δ / CI in 10^{-4} points.

Comp	2→3r1	3r1→3r2	3r2→3r3	3r3→4
S5E10	+0.103 [0.061,0.147] ✓	+0.346 [0.269,0.422] ✓	+0.063 [0.038,0.089] ✓	+0.086 [0.044,0.126] ✓

Comp	2→3r1	3r1→3r2	3r2→3r3	3r3→4
S6E1	+0.011 [-0.017,0.039] ✗	+2.041 [1.727,2.369] ✓	+0.620 [0.440,0.802] ✓	+1.235 [0.902,1.554] ✓
S6E2	+1.137 [0.917,1.352] ✓	+1.905 [1.628,2.174] ✓	+0.091 [0.027,0.155] ✓	+0.190 [0.102,0.277] ✓
S4E1	+0.004 [-0.072,0.081] ✗	+0.519 [0.088,0.966] ✓	-0.004 [-0.078,0.068] ✗	+0.391 [-0.259,1.080] ✗
S4E11	+0.569 [-1.066,1.493] ✗	+2.132 [-1.139,5.757] ✗	+1.422 [-1.208,4.336] ✗	+3.340 [-0.213,6.041] ✗

(✓ = CI excludes 0; ✗ = CI contains 0, i.e. inside OOF noise.)

- **S5E10** (largest, RMSE): every rung is significant — with 518k rows even a 6×10^{-6} RMSE step is resolvable.
- **S6E1 / S6E2** (630k): almost every rung significant; the exceptions are the first stage-3 round in S6E1 ($2 \rightarrow 3r1 = +1 \times 10^{-6} R^2$, $P=0.78$ — that round added essentially nothing; the real gains came in $r2/r3$) and a couple of marginal-but-still-significant steps.
- **S4E1** (165k): the ladder is carried almost entirely by a single round ($3r1 \rightarrow 3r2$). Two intra-stage-3 rounds are literal coin-flips ($P = 0.54$ and 0.46 ; one point- Δ is negative), and the tree-search step ($3r3 \rightarrow 4$) is inside noise. Only the accumulation survives.
- **S4E11** (141k, accuracy): **every individual rung is inside OOF noise** ($P = 0.63\text{--}0.97$) — yet the accumulated $2 \rightarrow 4$ is unambiguously significant ($P = 1.000$, CI $[3.1, 10.9] \times 10^{-4}$). This is the cleanest single illustration of the whole point: **a single 0.0x% step can be indistinguishable from OOF noise even when the end-to-end ladder is real.**

Honest calibration of the "the ladder holds in every competition" claim

- **What holds universally:** the **end-to-end** ladder gain (base blend $\rightarrow$ tree-search winner, $2 \rightarrow 4$) is statistically beyond OOF noise in **all five** competitions. That claim survives.
- **What does NOT hold universally:** the claim that each rung — in particular the tree-search rung ($3 \rightarrow 4$) — adds a resolvable improvement. In the two smaller competitions (S4E1, S4E11) that final rung is **inside OOF noise**, and several intra-stage-3 rounds are noise in S4E1/S4E11 (and one in S6E1). So "monotone improving ladder" is true as point estimates but several individual steps are not individually significant.
- **Even "significant" here is the weaker of the two senses of the word.** S4E1's $2 \rightarrow 4$ CI lower bound is $+3.7 \times 10^{-6}$ AUC — technically excluding 0, but so close that it would likely not survive a seed/fold re-run (which this analysis, by design, does not test). Read S4E1 as fragile-significant.
- **The two "inside noise" $3 \rightarrow 4$ verdicts are borderline, not decisive.** S4E1 $3 \rightarrow 4$ has $P=0.883$ and S4E11 $3 \rightarrow 4$ has $P=0.967$ (only 3.3% of resamples ≤ 0); both sit right on the 0 boundary and are method-sensitive — a normal-approx CI ($\pm 1.96 \cdot \text{SE}$) would in fact flip S4E11 $3 \rightarrow 4$ to significant, whereas the (skew-robust) percentile CI reported here keeps it inside noise. The honest reading is "not resolvable at the 95% percentile bar," not "proven zero." $B=5000$ is what makes even these calls stable; a coarse $B=200$ run flips them.

- **Consistent with the project's J-3 honest finding:** the gains are real in aggregate but small enough that individual micro-optimizations frequently sit within the estimator's own noise floor — the right posture is to trust the accumulated ladder, not to over-claim any single 0.0x% step.
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Reproduce & provenance

```
uv run python3 docs/scripts/bootstrap_ci.py          # human tables
uv run python3 docs/scripts/bootstrap_ci.py --json    # + machine-readable results
uv run python3 docs/scripts/bootstrap_ci.py --B 20000 # tighter CIs
```

- **Inputs (read-only):** competitions/playground-series-<comp>/scripts/cache/blend_{stage,tier}_{2,3_r1,3_r2,3_r3}.npz (stage 2/3 blend OOF); experiments_tree_v3.json → node_results (stage-4 winner members+weights); tree_search/cache_<comp>/solo_<id>.npz (member OOF for stage-4 reconstruction); data/train.csv (targets); experiments.json (recorded ladder values for the gate).
- **Scope covered:** S5E10, S6E1, S6E2 (primary) + S4E1, S4E11 (OOF caches found — included). No competition was skipped for missing OOF; only stage 1 (uncached Feb baseline) is omitted.
- **Method note:** percentile bootstrap CI, B = 5000, seed = 42 (reproducible; same seed per competition). Point Δ from full unresampled data.